

Fluorescent Dye Leak Detection



Fluorescent Dye Injection Using a R-134a Air Conditioning Service Unit or R-134a Air Conditioning Manifold Gauge Set and Fluorescent Dye Injector — Vehicles Requiring R-134a Addition

NOTE: This method of fluorescent dye injection requires the addition of R-134a from a R-134a Air Conditioning Service Unit or R-134a Air Conditioning Manifold Gauge Set hooked to an external tank to charge the dye into the refrigerant system. If adding fluorescent dye to a refrigerant system that is already fully charged, the Fluorescent Dye Injection Using a R-134a Fluorescent Dye Injector — Vehicles Not Requiring R-134a Addition method should be used.

NOTE: Fluorescent refrigerant system dye is added to the refrigerant system at the factory to assist in refrigerant system leak diagnosis using a Rotunda-approved UV lamp. It is not necessary to add additional dye to the refrigerant system before diagnosing leaks, even if a significant amount of refrigerant has been removed from the system. Replacement suction accumulators, receiver driers and receiver drier elements are shipped with a fluorescent dye wafer included in the desiccant bag which dissolves after approximately 30 minutes of continuous A/C operation. It is not necessary to add dye after flushing the refrigerant system because a new suction accumulator, receiver drier or receiver drier element is installed as part of the flushing procedure.

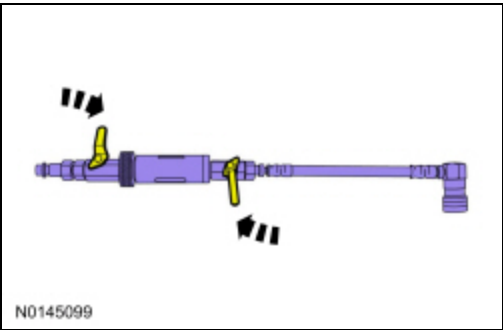
NOTE: Before using the R-134a Fluorescent Dye Injector for the first time, follow the equipment manufacturer procedures and instructions.

NOTE: Only connect the R-134a Fluorescent Dye Injector when fluorescent dye is to be injected. The dye injector has a one-way check valve that prevents refrigerant system recovery and evacuation.

1. **NOTE:** If no R-134a pressure is present in the refrigerant system, the system should be evacuated before carrying out the dye injection procedure. For additional information, refer to Air Conditioning (A/C) System Recovery, Evacuation and Charging.

Connect a R-134a Air Conditioning Service Unit or R-134a Air Conditioning Manifold Gauge Set to the refrigerant system service port valves.

2. Verify that the valves on the R-134a Fluorescent Dye Injector are closed.
- 219-703710 A/C Refrigerant R134A Loop/Add On Injector Kit-Set
- Use the General Equipment: Fluorescent Dye Injector



3. Fill the R-134a Fluorescent Dye Injector reservoir with 7.4 ml (0.25 fl. oz.) of fluorescent dye.



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4. Install the R-134a Fluorescent Dye Injector between the low-pressure service gauge port valve and the R-134a Air Conditioning Service Unit or R-134a Air Conditioning Manifold Gauge Set.

5. **NOTE:** *Following fluorescent dye injection, the refrigerant system should be fully charged to make sure of correct movement of the dye. The dye moves with the oil not the refrigerant. The oil moves with the refrigerant.*

Open all valves and inject the fluorescent dye into the refrigerant system by charging the refrigerant system with the required amount of R-134a.

6. When fluorescent dye injection is complete, close all valves.

7. Recover the refrigerant from the dye/lubricant injector.

8. Remove the dye/lubricant injector from the low-pressure service gauge port valve and the R-134a Air Conditioning Service Unit or R-134a Air Conditioning Manifold Gauge Set.

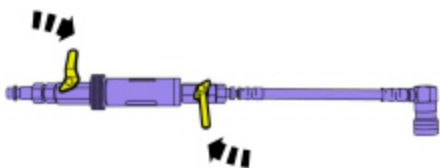
Fluorescent Dye Injection Using a R-134a Fluorescent Dye Injector — Vehicles Not Requiring R-134a Addition

NOTE: *Fluorescent refrigerant system dye is added to the refrigerant system at the factory to assist in refrigerant system leak diagnosis using a Rotunda-approved UV lamp. It is not necessary to add additional dye to the refrigerant system before diagnosing leaks, even if a significant amount of refrigerant has been removed from the system. Replacement suction accumulators, receiver driers and receiver drier elements are shipped with a fluorescent dye wafer included in the desiccant bag which dissolves after approximately 30 minutes of continuous A/C operation. It is not necessary to add dye after flushing the refrigerant system because a new suction accumulator, receiver drier or receiver drier element is installed as part of the flushing procedure.*

NOTE: *Before using the R-134a Fluorescent Dye Injector for the first time, follow the equipment manufacturer procedures and instructions.*

NOTE: *Refrigerant system pressure should be between 413-551 kPa (60-80 psi) at 24°C (75°F) with the engine off and cool.*

1. Verify that the valves on the R-134a Fluorescent Dye Injector are closed.
 - 219-703710 A/C Refrigerant R134A Loop/Add On Injector Kit-Set
 - Use the General Equipment: Fluorescent Dye Injector



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2. Fill the R-134a Fluorescent Dye Injector reservoir with 7.4 ml (0.25 fl. oz.) of fluorescent dye.



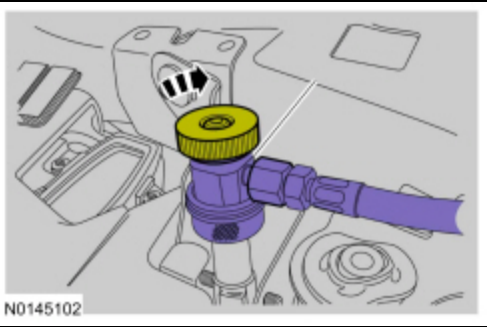
3. Install the R-134a Fluorescent Dye Injector between the high-pressure and low-pressure service gauge port valves.

4. **NOTICE:** Make sure all tools and hoses are clear of the engine cooling fan and drive belt before starting the engine. Failure to keep tools and hoses clear from the engine cooling fan and drive belt will result in damage to the tools and/or vehicle.

With the A/C off, start the engine. Allow engine speed to stabilize below 1,000 rpm.

5. Set the A/C to the ON position.

6. Open the high-pressure service valve.

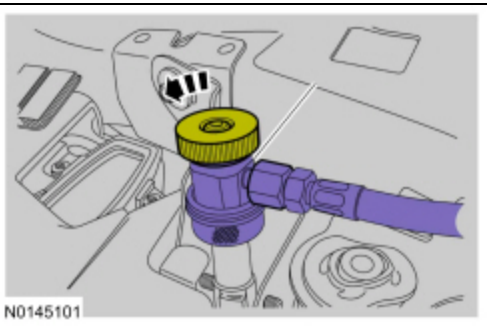


7. **NOTE:** To prevent pressure spike/liquid slug, slowly open the R-134a Fluorescent Dye Injector valves and inject the fluorescent dye into the refrigerant system.

Open the R-134a Fluorescent Dye Injector valves and inject the fluorescent dye into the refrigerant system.



8. Close the high-pressure service valve to allow the pressure inside the R-134a Fluorescent Dye Injector to equalize with the suction side of the refrigerant system.



9. **NOTE:** Close the valves on the R-134a Fluorescent Dye Injector while the A/C compressor is operating.

Close the valves on the R-134a Fluorescent Dye Injector.



10. **NOTE:** *Make sure all of the valves on the R-134a Fluorescent Dye Injector remain closed when not in use.*

Disconnect the high-pressure and low-pressure service valves and remove the R-134a Fluorescent Dye Injector from the vehicle.

Fluorescent Dye Detection

NOTE: *Ford Motor Company vehicles are produced with R-134a fluorescent dye installed in the refrigerant system from the factory. The location of leaks can be pinpointed by the bright yellow-green glow of the fluorescent dye under a Rotunda-approved UV lamp. Since more than one leak can exist, make sure to inspect each component, line and fitting in the refrigerant system for a leak.*

NOTE: *Use of dye-enhancing glasses or goggles greatly improves the detection of the dye under the Rotunda-approved UV lamp.*

NOTE: *Not all UV lamps will fluoresce the dye used in Ford vehicles. All Rotunda UV lamps are optimized to fluoresce the dye.*

1. Check for leaks using a Rotunda-approved UV lamp and dye enhancing glasses.
2. Inspect all components, lines and fittings of the refrigerant system.
3. After the leak(s) is repaired, remove any traces of fluorescent dye with a general purpose oil solvent.
4. Verify the repair by running the vehicle for a short period of time and rechecking the area of the leak with a Rotunda-approved UV lamp.
 - UVU413025 UV Leak Detection Light U-View
Use the General Equipment: UV Leak Detector